

Azure Databricks: Build a Trusted Data Foundation with Medallion Architecture

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My experiences includes:

- Avanade
- CAE
- Air Canada



Expected Audience and Presentation Structure



What We'll Cover:

- Sign-up for a free Databricks account.
- Introduction to Medallion Architecture and why it is used?
- Understanding of Batch vs Streaming data pipeline.
- Live Demo: Build a Medallion Architecture with Structured Streaming using Yelp Open Dataset.



Fundamental knowledge:

- Basic familiarity with Databricks and PySpark.
- Understanding of basic data engineering concepts and common data quality issues is beneficial, but not required.

Databricks Free Edition: What it is & When to use it?

What it can do

- Sandbox for learning or personal projects
- Same tool as the enterprise version with some limited capabilities
- Easy to start: Sign-up and go.

Link: <https://learn.microsoft.com/en-us/azure/databricks/getting-started/free-edition>

Limitations

- Limited computing power
- One Databricks workspace per account
- Data is stored in the Databricks default storage
- Inactive accounts for a prolonged period might be deleted
- No security customization
- No SLA (its free!)

To really learn the platform, we need to work with real data

Yelp Open Dataset

A real-world dataset designed for educational use.

yelp  data licensing



6,990,280

Reviews



150,346

Businesses



11

Metropolitan areas

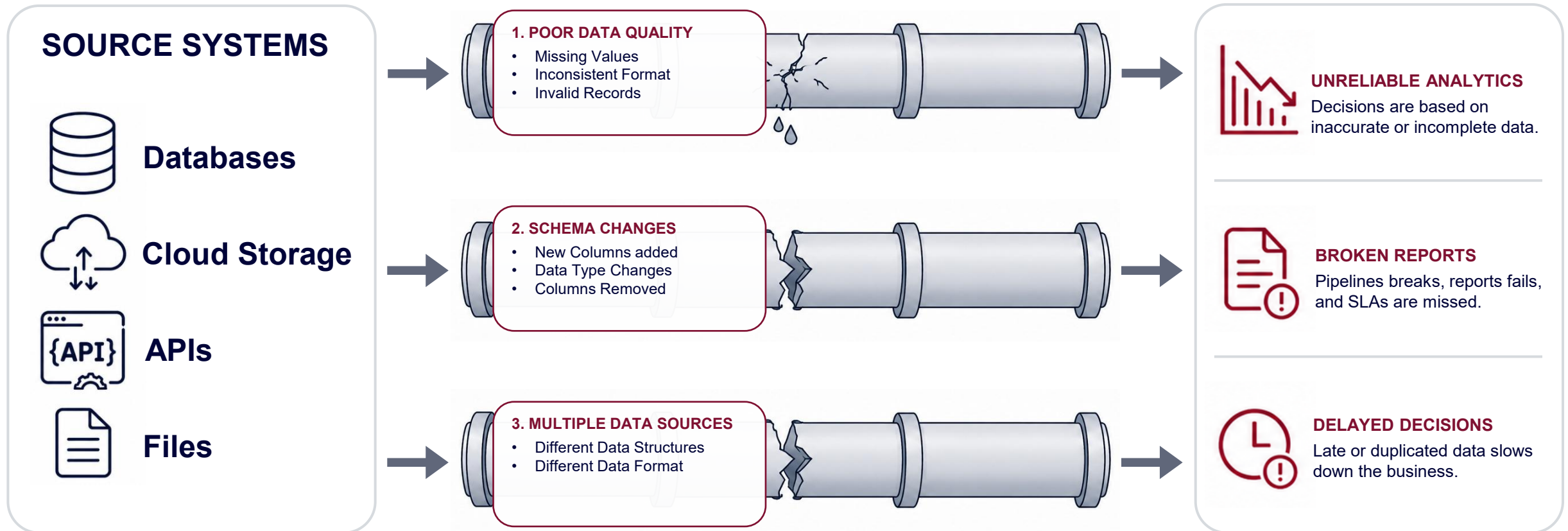


200,100

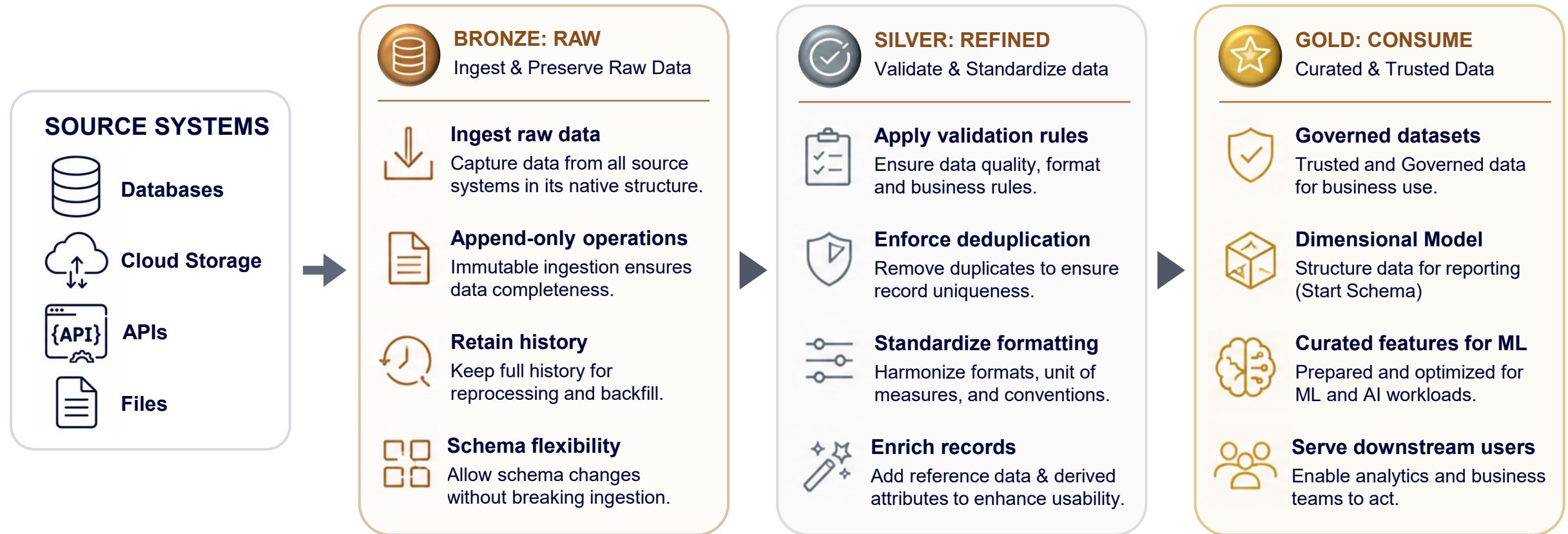
Pictures

Link: <https://business.yelp.com/data/resources/open-dataset/>

Without a systematic approach to data processing, source system issues cascade through pipelines and undermine data trust.



Medallion Architecture provides a structured framework to incrementally validate and refine data for downstream consumption.



Medallion Architecture can be implemented using batch and streaming approaches.



Batch Processing

Processes data on a scheduled interval (often daily or twice a day).

Best suited for historical reporting with lower-frequency updates.



Structured Streaming

Processes data continuously as new events arrives. Supports both event-drive and micro-batch execution.

Best suited for near real-time reporting or high-frequency updates.



Lakeflow Declarative Pipelines (SDP)

Builds and manages data pipelines using a configuration-driven approach, reducing the amount of code required.

It can be used in both Batch or Streaming scenarios.



We are going to focus on Structured Streaming

1. Near Real-time processing

Provides visibility into how data flows through the Medallion Architecture in micro-batching.

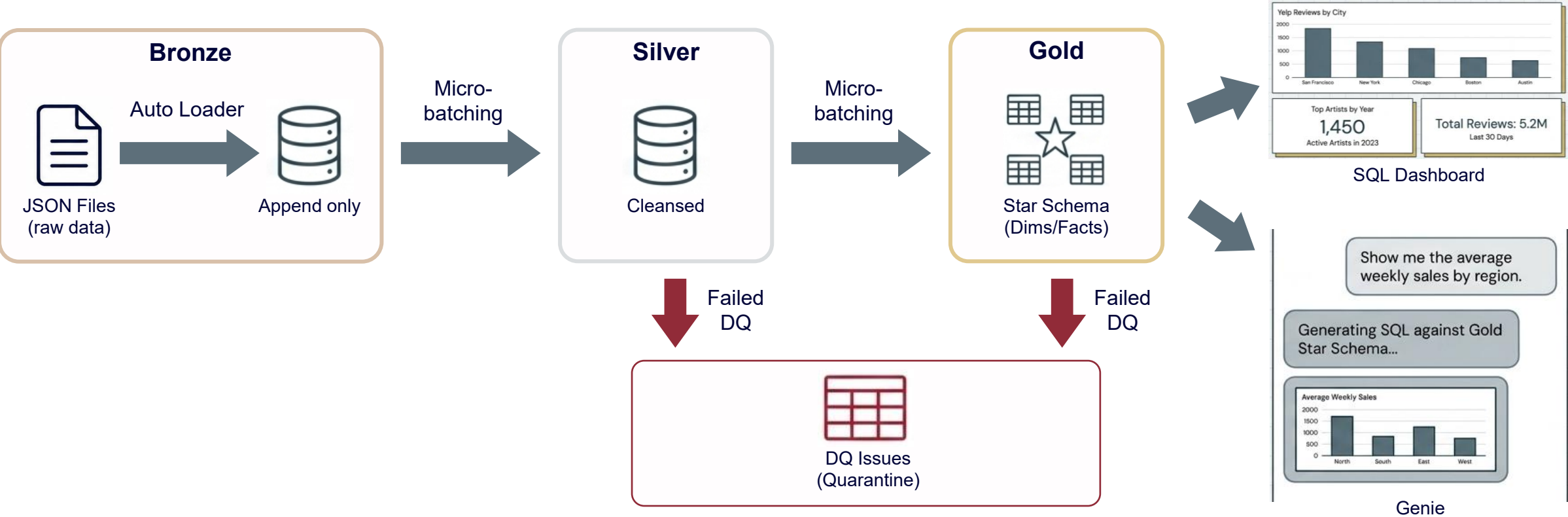
2. Data quality in motion

Demonstrates how data quality rules are applied and handled.

3. Foundation for SDP

Establishes core concepts that underpin the Lakeflow Declarative Pipelines

Demo: End-to-End data pipeline using Structured Streaming

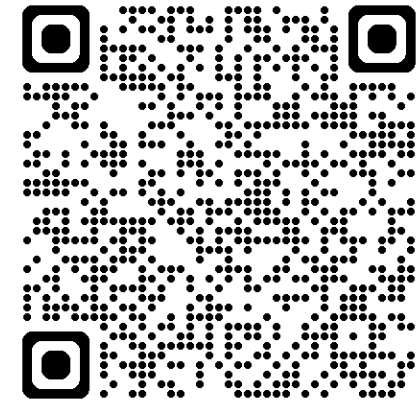


Building Trusted Data Pipeline with Medallion Architecture

Key Takeaways:

- Databricks Free Edition lets you experiment and learn using real-work data and tool.
- Medallion Architecture provides a structured framework to progressively improve data quality and trust through the Bronze, Silver and Gold layers.
- The same Medallion Architecture can be implemented using Batch, Structured Streaming and Lakeflow Declarative Pipelines (SDP).
- Structured Streaming provides the core concepts to help us understand SDP.

Resources



GitHub Repo

<https://github.com/wilson-mok/demo/tree/main/2026/EDS/2-Medallion-Architecture-Structured-Streaming>

Thank you



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